

Homework 1: DSP Review

1 Basic Properties of Linear-Phase Filters

- (a) Recall from lecture that Type II filters cannot be high-pass. Determine (with explanations) which kinds of filters are realizable by Type I, II, III, and IV filters from the list: all-pass, band-pass, low-pass, band-cut, high-pass, all-cut.

Solution.

- (a) **Type-I Filters** have an *amplitude response* of the form

$$H_{\text{ampl}}(\omega) = \sum_{n=0}^L a[n] \cos(n\omega). \quad (1.1)$$

The problem then boils down to understanding when an amplitude response of this form is necessarily zero, where we can take $a[n]$ to be an arbitrary coefficient sequence. Observe that there are no frequencies ω where H_{ampl} is necessarily zero so Type-I filters can implement any kind filter except for an all-pass filter (other than the pure delay all-pass filter). Note that the all-cut filter is trivially implemented by taking $a[n]$ to be the 0 sequence.

- (b) **Type-II Filters:** have an *amplitude response* of the form

$$H_{\text{ampl}}(\omega) = \sum_{n=0}^L b[n] \cos((n + 1/2)\omega). \quad (1.2)$$

The problem then boils down to understanding when an amplitude response of this form is necessarily zero, where we can take $b[n]$ to be an arbitrary coefficient sequence. Observe that this form necessarily has a zero at $\omega = \pi$, so Type-II filters can only implement low-pass, band-pass, and all-cut filters. Note that the all-cut filter is trivially implemented by taking $b[n]$ to be the 0 sequence.

- (c) **Type-III Filters:** have an *amplitude response* of the form

$$H_{\text{ampl}}(\omega) = \sum_{n=0}^L c[n] \sin(n\omega). \quad (1.3)$$

The problem then boils down to understanding when an amplitude response of this form is necessarily zero, where we can take $c[n]$ to be an arbitrary coefficient sequence. Observe that this form necessarily has a zero at $\omega = 0$ and $\omega = \pi$, so Type-III filters can only implement band-pass or all-cut filters. Note that the all-cut filter is trivially implemented by taking $c[n]$ to be the 0 sequence.

- (d) **Type-IV Filters:** have an *amplitude response* of the form

$$H_{\text{ampl}}(\omega) = \sum_{n=0}^L d[n] \sin((n + 1/2)\omega). \quad (1.4)$$

The problem then boils down to understanding when an amplitude response of this form is necessarily zero, where we can take $d[n]$ to be an arbitrary coefficient sequence. Observe that this form necessarily has a zero at $\omega = 0$, so Type-IV filters can only implement high-pass, band-pass, or all-cut filters. Note that the all-cut filter is trivially implemented by taking $d[n]$ to be the 0 sequence.

- (b) Can IIR filters be linear-phase? Explain why or why not.

Solution. Many people like to claim that linear-phase filters cannot be IIR, but this concept is actually more subtle. Indeed, recall that linear-phase filters (with real coefficients) are either symmetric or antisymmetric. It turns out that IIR filters that are either symmetric or antisymmetric are also linear-phase.

For example, consider the symmetric sequence with center at $L = 0$ specified by $h[n] = e^{-|n|}$. The z -transform of this sequence is

$$H(z) = \sum_{n=-\infty}^{\infty} e^{-|n|} z^{-n} = 1 + \frac{e^{-1} z^{-1}}{1 - e^{-1} z^{-1}} + \frac{e^{-1} z}{1 - e^{-1} z}, \quad (1.5)$$

which has poles at $z = e^{-1}$ and $z = e$. The corresponding region of convergence is $e^{-1} < |z| < e$, which contains the unit circle, so the frequency response of the system is well-defined. Evaluating on the unit circle $z = e^{j\omega}$ gives

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} e^{-|n|} e^{-j\omega n} = \frac{1 - e^{-2}}{1 - 2e^{-1} \cos \omega + e^{-2}}, \quad (1.6)$$

which is a real and nonnegative expression. Thus the phase response is identically zero, confirming that $h[n] = e^{-|n|}$ defines a noncausal IIR filter with linear (zero) phase. However, since it is *noncausal*. Causal IIR filters cannot be linear phase (other than the trivial (i.e., all-cut) filter with all coefficients equal to 0).

2 Structure of Linear-Phase Filters

- (a) Consider a linear-phase filter $h[n]$ with **real coefficients** and suppose that z_0 is a zero of $H(z)$, i.e., $H(z_0) = 0$. Prove that this constraint enforces that $(z_0, z_0^{-1}, z_0^*, z_0^{-1*})$ are all zeros of $H(z)$. *Hint: Linear-phase filters with real coefficients satisfy $h[n] = \pm h[2L - n]$, where L is the center of the filter.*

Solution. If we take the z -transform of the identity $h[n] = \pm h[2L - n]$ we find

$$H(z) = \pm z^{-2L} H(z^{-1}). \quad (2.1)$$

Since $h[n]$ has real coefficients, we also have the identity

$$H(z) = H^*(z^*). \quad (2.2)$$

This then implies that we also have that

$$H(z) = \pm z^{-2L} H^*(z^{-1*}). \quad (2.3)$$

Thus, if $H(z_0) = 0$, these three equations immediately imply that z_0^{-1} , z_0^* , and z_0^{-1*} are also zeros.

- (b) Recall from lecture that Type I and II filters with real coefficients are **symmetric** and satisfy $h[n] = h[2L - n]$ and similarly Type III and IV filters with real coefficients are *antisymmetric* and satisfy $h[n] = -h[2L - n]$. When linear-phase filters have **complex coefficients**, the impulse responses should be **conjugate-symmetric** or **conjugate-antisymmetric**.

- (i) Write down the identities that complex-coefficient linear-phase filters must satisfy. Explain why these identities reduce to the ones discussed in lecture for real-coefficient linear-phase filters.

Solution. Type I and II filters satisfy

$$h[n] = h^*[2L - n]. \quad (2.4)$$

Type III and IV filters satisfy

$$h[n] = -h^*[2L - n]. \quad (2.5)$$

When $h[n]$ has real coefficients $h^*[2L - n] = h[2L - n]$ and so the identities reduce to the ones discussed in lecture.

- (ii) If $h[n]$ is a complex-coefficient linear-phase filter and z_0 is a zero of $H(z)$, for what other values is $H(z)$ necessarily zero?

Solution. We have the identity $h[n] = \pm h^*[2L - n]$. Taking the z -transform we find

$$H(z) = z^{-2L} H^*(z^{-1*}). \quad (2.6)$$

This if $H(z_0) = 0$, then z_0^{-1*} must be zero.

- (c) Suppose $re^{j\theta}$ is a zero of a linear-phase filter with real coefficients. Determine $H(z)$ so that it is linear-phase with real coefficients.

Solution. There are many filters that satisfy this property. Writing $z_0 = re^{j\theta}$, the general form that captures all solutions is the following:

$$H(z) = Kz^{-L}(1 - z_0z^{-1})(1 - z_0^*z^{-1})(1 - z_0^{-1}z^{-1})(1 - z_0^{-*}z^{-1})G(z), \quad (2.7)$$

where z^{-L} is a pure delay, K is a real number, and $G(z)$ is any product of the four factors $(1 - z_0z^{-1})$, $(1 - z_0^*z^{-1})$, $(1 - z_0^{-1}z^{-1})$, $(1 - z_0^{-*}z^{-1})$, which need to be included explicitly in the form of $H(z)$ so that we have that the four-tuple of zeros that are required of linear-phase filters with real coefficients.

3 Minimum-Phase and Maximum-Phase Systems

Consider the system defined by the input-output relation

$$y[n] = 4x[n] + 4x[n - 1] + 4y[n - 2]$$

- (a) Compute the transfer function $H(z)$ of this system and draw its pole-zero plot.

Solution. In the z -domain, this system corresponds to the following equation:

$$Y(z) = 4X(z) + 4X(z)z^{-1} + 4Y(z)z^{-2}. \quad (3.1)$$

Therefore,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{4 + 4z^{-1}}{1 - 4z^{-2}} = \frac{4 + 4z^{-1}}{(1 - 2z^{-1})(1 - 2z^{-1})}. \quad (3.2)$$

Therefore, poles are $z = \pm 2$, and zeros are $z = 0$ and $z = -1$. A pole-zero plot is in Fig. 1.

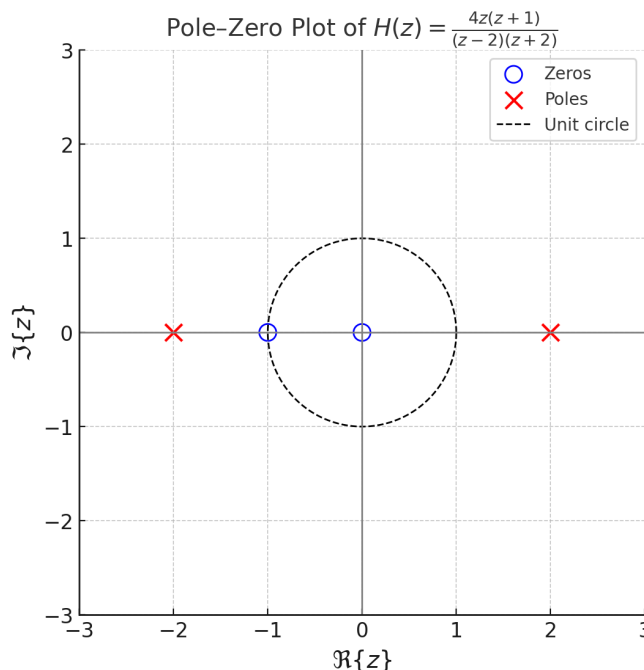


Fig. 1: Pole-zero plot of $H(z)$.

- (b) Determine the minimum-phase transfer function $H_{\min}(z)$ and draw its pole-zero plot.

Solution. Since the original system is unstable, the notion of a minimum-phase system is not well-defined. If one were to be pedantic, in this case we can apply the algorithmic trick and flip all zeros to be inside the unit circle, but since the original system already has all zeros inside (or on) the unit circle, one could also argue that $H_{\min}(z) = H(z)$.

- (c) Determine the maximum-phase transfer function $H_{\max}(z)$ and draw its pole-zero plot.

Solution. Since the original system is unstable, the notion of a maximum-phase system is not well-defined. If one were to be pedantic, this would still be the case. Indeed, to find the maximum-phase system, the algorithmic trick is to flip all zeros to be outside the unit circle. In this case, the zero at 0 would be flipped to be at ∞ . If you try to do this naively from the transfer function, you cannot realize a zero at ∞ .